

BG77xA-GL&BG95xA-GL

QNWCFG AT Commands Manual

LPWA Module Series

Version: 1.0

Date: 2024-05-31

Status: Released



At Quectel, our aim is to provide timely and comprehensive services to our customers. If you require any assistance, please contact our headquarters:

Quectel Wireless Solutions Co., Ltd.

Building 5, Shanghai Business Park Phase III (Area B), No.1016 Tianlin Road, Minhang District, Shanghai 200233, China

Tel: +86 21 5108 6236

Email: info@quectel.com

Or our local offices. For more information, please visit:

<http://www.quectel.com/support/sales.htm>.

For technical support, or to report documentation errors, please visit:

<http://www.quectel.com/support/technical.htm>.

Or email us at: support@quectel.com.

Legal Notices

We offer information as a service to you. The provided information is based on your requirements and we make every effort to ensure its quality. You agree that you are responsible for using independent analysis and evaluation in designing intended products, and we provide reference designs for illustrative purposes only. Before using any hardware, software or service guided by this document, please read this notice carefully. Even though we employ commercially reasonable efforts to provide the best possible experience, you hereby acknowledge and agree that this document and related services hereunder are provided to you on an “as available” basis. We may revise or restate this document from time to time at our sole discretion without any prior notice to you.

Use and Disclosure Restrictions

License Agreements

Documents and information provided by us shall be kept confidential, unless specific permission is granted. They shall not be accessed or used for any purpose except as expressly provided herein.

Copyright

Our and third-party products hereunder may contain copyrighted material. Such copyrighted material shall not be copied, reproduced, distributed, merged, published, translated, or modified without prior written consent. We and the third party have exclusive rights over copyrighted material. No license shall be granted or conveyed under any patents, copyrights, trademarks, or service mark rights. To avoid ambiguities, purchasing in any form cannot be deemed as granting a license other than the normal non-exclusive, royalty-free license to use the material. We reserve the right to take legal action for noncompliance with abovementioned requirements, unauthorized use, or other illegal or malicious use of the material.

Trademarks

Except as otherwise set forth herein, nothing in this document shall be construed as conferring any rights to use any trademark, trade name or name, abbreviation, or counterfeit product thereof owned by Quectel or any third party in advertising, publicity, or other aspects.

Third-Party Rights

This document may refer to hardware, software and/or documentation owned by one or more third parties ("third-party materials"). Use of such third-party materials shall be governed by all restrictions and obligations applicable thereto.

We make no warranty or representation, either express or implied, regarding the third-party materials, including but not limited to any implied or statutory, warranties of merchantability or fitness for a particular purpose, quiet enjoyment, system integration, information accuracy, and non-infringement of any third-party intellectual property rights with regard to the licensed technology or use thereof. Nothing herein constitutes a representation or warranty by us to either develop, enhance, modify, distribute, market, sell, offer for sale, or otherwise maintain production of any our products or any other hardware, software, device, tool, information, or product. We moreover disclaim any and all warranties arising from the course of dealing or usage of trade.

Privacy Policy

To implement module functionality, certain device data are uploaded to Quectel's or third-party's servers, including carriers, chipset suppliers or customer-designated servers. Quectel, strictly abiding by the relevant laws and regulations, shall retain, use, disclose or otherwise process relevant data for the purpose of performing the service only or as permitted by applicable laws. Before data interaction with third parties, please be informed of their privacy and data security policy.

Disclaimer

- a) We acknowledge no liability for any injury or damage arising from the reliance upon the information.
- b) We shall bear no liability resulting from any inaccuracies or omissions, or from the use of the information contained herein.
- c) While we have made every effort to ensure that the functions and features under development are free from errors, it is possible that they could contain errors, inaccuracies, and omissions. Unless otherwise provided by valid agreement, we make no warranties of any kind, either implied or express, and exclude all liability for any loss or damage suffered in connection with the use of features and functions under development, to the maximum extent permitted by law, regardless of whether such loss or damage may have been foreseeable.
- d) We are not responsible for the accessibility, safety, accuracy, availability, legality, or completeness of information, advertising, commercial offers, products, services, and materials on third-party websites and third-party resources.

Copyright © Quectel Wireless Solutions Co., Ltd. 2024. All rights reserved.

About the Document

Revision History

Version	Date	Author	Description
-	2024-03-29	Nikola Radosavljevic	Creation of the document
1.0	2024-05-31	Nikola Radosavljevic	First official release

Contents

About the Document.....	3
Contents	1
Table Index.....	5
1 Introduction	6
2 AT Command Introduction	7
2.1. Definitions	7
2.2. AT Command Syntax.....	7
2.3. AT Command Responses.....	8
2.4. Declaration of AT Command Examples	8
3 Description of AT+QNWCFG Commands	10
3.1. AT+QNWCFG Extended Configuration Settings	10
3.1.1. AT+QNWCFG="backoff" Query Remaining Back-off Timer Value for All Rejecting PLMNs.....	10
3.1.2. AT+QNWCFG="ta" Query Timing Advance Value Under Current RAT	11
3.1.3. AT+QNWCFG="cfun_init_mode" Set CFUN Booting Mode	12
3.1.4. AT+QNWCFG="cmatt" Attach to or Detach from LTE Network	13
4 Summary of CME Error Codes	14
5 Appendix References	16

Table Index

Table 1: Applicable Modules 6

Table 2: Types of AT Commands 7

Table 3: Summary of General +CME ERROR Codes 14

Table 4: Related Documents..... 16

Table 5: Terms and Abbreviations 16

1 Introduction

This document describes the **AT+QNWCFG** commands supported on Quectel BG77xA-GL and BG95xA-GL modules.

Table 1: Applicable Modules

Module Series	Model
BG77xA-GL	BG770A-GL
	BG772A-GL
	BG773A-GL
BG95xA-GL	BG950A-GL
	BG951A-GL
	BG952A-GL
	BG953A-GL
	BG955A-GL

2 AT Command Introduction

2.1. Definitions

- **<CR>** Carriage return character.
- **<LF>** Line feed character.
- **<...>** Parameter name. Angle brackets do not appear on the command line.
- **[...]** Optional parameter of a command or an optional part of TA information response. Square brackets do not appear on the command line. When an optional parameter is not given in a command, the new value equals its previous value or the default settings, unless otherwise specified.
- **Underline** Default setting of a parameter.

2.2. AT Command Syntax

All command lines must start with **AT** or **at** and end with **<CR>**. Information responses and result codes always start and end with a carriage return character and a line feed character: **<CR><LF><response><CR><LF>**. In tables presenting commands and responses throughout this document, only the commands and responses are presented, and **<CR>** and **<LF>** are deliberately omitted.

AT+QNWCFG commands implemented by the module are in “Extended” syntax, as illustrated below.

- **Extended**

There are several types of extended commands as shown in the following table.

Table 2: Types of AT Commands

Command Type	Syntax	Description
Test Command	AT+<cmd>=?	Test the existence of the corresponding command and return information about the type, value, or range of its

		parameter.
Read Command	AT+<cmd>?	Check the current parameter value of the corresponding command.
Write Command	AT+<cmd>=<p1>[,<p2>[,<p3>[...]]]	Set user-definable parameter value.
Execution Command	AT+<cmd>	Return a specific information parameter or perform a specific action.

Multiple commands can be placed on a single line using a semi-colon (;) between commands. In such cases, only the first command should have **AT** prefix. Commands can be in upper or lower case.

Spaces should be ignored when you enter AT commands, except in the following cases:

- Within quoted strings, where spaces are preserved;
- Within an unquoted string or numeric parameter;
- Within an IP address;
- Within the AT command name up to and including a =, ? or =?.

On input, at least a carriage return is required. A newline character is ignored so it is permissible to use carriage return/line feed pairs on the input.

If no command is entered after the **AT** token, **OK** will be returned. If an invalid command is entered, **ERROR** will be returned.

Optional parameters, unless explicitly stated, need to be provided up to the last entered parameter.

2.3. AT Command Responses

When the AT command processor has finished processing a line, it will output **OK**, **ERROR** or **+CME ERROR: <err>** to indicate that it is ready to accept a new command. Solicited information responses are sent before the final **OK**, **ERROR** or **+CME ERROR: <err>**.

Responses will be in the format of:

```
<CR><LF>+CMD1:<parameters><CR><LF>
<CR><LF>OK<CR><LF>
```

2.4. Declaration of AT Command Examples

The AT command examples in this document are provided to help you learn about the use of the AT

commands introduced herein. The examples, however, should not be taken as Quectel's recommendations or suggestions about how to design a program flow or what status to set the module into. Sometimes multiple examples may be provided for one AT command. However, this does not mean that there is a correlation among these examples, or that they should be executed in a given sequence.

3 Description of AT+QNWCFG Commands

3.1. AT+QNWCFG Extended Configuration Settings

The following Test Command shows the supported configuration settings of the module.

AT+QNWCFG Extended Configuration Settings	
Test Command AT+QNWCFG=?	Response +QNWCFG: "backoff" +QNWCFG: "ta" +QNWCFG: "cfun_init_mode",(list of supported <mode>s) +QNWCFG: "cmatt",(list of supported <action>s) OK
Maximum Response Time	300 ms
Characteristics	-

3.1.1. AT+QNWCFG="backoff" Query Remaining Back-off Timer Value for All Rejecting PLMNs

This command queries the remaining back-off timer value for all PLMNs that rejected a connection request and set a back-off timer.

AT+QNWCFG="backoff" Query Remaining Back-off Timer Value for All Rejecting PLMNs	
Write Command AT+QNWCFG="backoff"	Response [+QNWCFG: "backoff",<info1>,<info2>,<info3>,<info4> [+QNWCFG: "backoff",<info1>,<info2>,<info3>,<info4> [...]]] OK If there is an error related to ME functionality:

	+CME ERROR: <err> If there is any other error: ERROR
Maximum Response Time	300 ms
Characteristics	-

Parameter

<info1>	Integer type. Back-off timer status. 0 Stop 1 Running
<info2>	Integer type. Remaining back-off time. 0 or omitted if the timer is stopped. Unit: ms.
<info3>	Integer type. PLMN.
<info4>	Integer type. CID (Context ID associated with the PLMN).
<err>	Error code. For more details, see Chapter 4 .

3.1.2. AT+QNWCFG="ta" Query Timing Advance Value Under Current RAT

This command queries the timing advance value under the current RAT.

AT+QNWCFG="ta" Query Timing Advance Value Under Current RAT	
Write Command AT+QNWCFG="ta"	Response +QNWCFG: "ta",<current_RAT>,<ta_value> OK If there is an error related to ME functionality: +CME ERROR: <err> If there is any other error: ERROR
Maximum Response Time	300 ms
Characteristics	-

Parameter

<current_RAT>	String type. Current radio access technology that the module is connected to.
"eMTC"	Cat M1
"NBIoT"	Cat NB1/NB2

<ta_value>	Integer type. Timing Advance value (N_{TA}) of measured cell is represented by index values of $T_A = 0, 1, 2, \dots, 1282$. The N_{TA} value corresponds to the time alignment and is calculated as $N_{TA} = T_A \times 16$ per 3GPP TS 36.213.
<err>	Error code. For more details, see Chapter 4 .

NOTE

BG955A-GL module supports GSM, but this RAT is not supported for this command.

3.1.3. AT+QNWCFG="cfun_init_mode" Set CFUN Booting Mode

This command sets the CFUN booting mode of the module.

AT+QNWCFG="cfun_init_mode" Set CFUN Booting Mode	
Write Command AT+QNWCFG="cfun_init_mode" [,<mode>]	Response If the optional parameter is omitted, query the current setting: +QNWCFG: "cfun_init_mode",<mode> OK If the optional parameter is specified, set the CFUN booting mode: OK If there is an error related to ME functionality: +CME ERROR: <err> If there is any other error: ERROR
Maximum Response Time	300 ms
Characteristics	The command takes effect after the module reboots. The configuration is saved automatically.

Parameter

<mode>	Integer type. CFUN booting mode for the module. 0 Limited functionality <u>1</u> Full functionality
<err>	Error code. For more details, see Chapter 4 .

Example

AT+QNWCFG="cfun_init_mode",0 //Set the module CFUN booting mode to limited functionality.

```
OK
AT+QNWCFG="cfun_init_mode" //Query the current setting.
+QNWCFG: "cfun_init_mode",0
OK
```

NOTE

When **<mode>** is 0, the actual state of the module is the same as that of **AT+CFUN=4**.

3.1.4. AT+QNWCFG="cmatt" Attach to or Detach from LTE Network

This command sets UE functionality and attaches to or detaches from the LTE network.

AT+QNWCFG="cmatt" Attach to or Detach from LTE Network

Write Command AT+QNWCFG="cmatt"[,<action>]	Response If the optional parameter is omitted, query the current setting: +QNWCFG: "cmatt",<action> OK If the optional parameter is specified, attach to or detach from the LTE network. OK If there is an error related to ME functionality: +CME ERROR: <err> If there is any other error: ERROR
Maximum Response Time	300 ms
Characteristics	The command takes effect after the module reboots. The configurations are saved automatically.

Parameter

<action>	Integer type. Attach to or detach from the LTE network. 0 Detach <u>1</u> Set full UE functionality (with RF front-end and SIM card enabled) and attach
<err>	Error code. For more details, see Chapter 4 .

4 Summary of CME Error Codes

Final result code **+CME ERROR: <err>** indicates an error related to mobile equipment or network. **+CME ERROR: <err>** final result code functions the same as the regular **ERROR** result code: if **+CME ERROR: <err>** is the result code for any of the commands in a command line, none of the following commands in the same command line is executed (neither **ERROR** nor **OK** result code shall be returned as a result of a completed command line execution). The format of **<err>** can be either numeric or verbose. This is set with **AT+CMEE**. For more details, see [document \[1\]](#).

Table 3: Summary of General +CME ERROR Codes

Numeric <err> Value	Verbose <err> Value
0	phone failure
1	no connection to phone
2	phone-adaptor link reserved
3	operation not allowed
4	operation not supported
5	PH-SIM PIN required
6	PH-FSIM PIN required
7	PH-FSIM PUK required
10	SIM not inserted
11	SIM PIN required
12	SIM PUK required
13	SIM failure
14	SIM busy
15	SIM wrong
16	incorrect password

17	SIM PIN2 required
18	SIM PUK2 required
20	memory full
21	invalid index
22	not found
23	memory failure
24	text string too long
25	invalid characters in text string
26	dial string too long
27	invalid characters in dial string
30	no network service
31	network timeout
32	network not allowed - emergency calls only
40	network personalization PIN required
41	network personalization PUK required
42	network subset personalization PIN required
43	network subset personalization PUK required
44	service provider personalization PIN required
45	service provider personalization PUK required
46	corporate personalization PIN required
47	corporate personalization PUK required
50	Incorrect parameters

5 Appendix References

Table 4: Related Documents

Document Name
[1] Quectel_BG77xA-GL&BG95xA-GL&BG950S-GL_AT_Commands_Manual

Table 5: Terms and Abbreviations

Abbreviation	Description
AT	Attention
eMTC	enhanced Machine-Type Communication
PLMN	Public Land Mobile Network
GSM	Global System for Mobile Communications
ID	Identifier
ME	Mobile Equipment
NB-IoT	Narrowband Internet of Things
PIN	Personal Identification Number
PUK	PIN Unlock Key
RAT	Radio Access Technology
TA	Terminal Adapter
UE	User Equipment